

Software Requirements Specification

Travel By MoMo — Internal Travel & Expense Platform (Corporate Company)

Version 0.3 (Draft)

Revision: adds Mobile Money (MoMo) as the primary payment method (Sections 3.6, 4.3)

Prepared for internal engineering scoping

August 28, 2026

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for Travel By MoMo, a multi-tenant platform that lets employees at each client company search, book, get approval for, pay for, and expense business travel in one workflow, using Mobile Money as the primary payment method. It is intended to give engineering a concrete basis for design and estimation, and to give stakeholders (Finance, Travel Admin, HR) a way to confirm the system will do what they need before build starts.

1.2 Intended Audience and Use

- Engineering — as the basis for technical design, sprint planning, and acceptance criteria
- Finance / Travel Admin — to confirm policy, payment, and reporting needs are captured
- HR / People Ops — to confirm duty-of-care and traveler-data requirements are captured
- Project sponsor — as a scope reference for what is (and isn't) in each phase

1.3 Product Scope

Travel By MoMo is not a replacement for a GDS (Global Distribution System) or for a mobile money network. It is a multi-tenant workflow layer that:

- Sources flight/hotel content and booking capability from a licensed third-party booking API
- Adds tenant-specific policy rules, approval routing, payment control, and expense automation on top
- Uses Mobile Money (MTN MoMo, Airtel Money) as the primary, traveler-facing payment rail, with virtual cards used only as a settlement bridge to suppliers that require international card-network payment
- Serves multiple client companies (tenants) from a single shared platform, each fully isolated from the others

Out of scope for all phases:

- Direct airline/hotel contracting
- GDS/NDC accreditation
- Becoming an IATA-accredited travel agency
- Public/external customer use outside of onboarded tenant companies
- Becoming a licensed mobile money operator or payment service provider — the system integrates with existing MoMo networks/aggregators rather than replacing them

1.4 Definitions, Acronyms, and Abbreviations

Term	Meaning
PNR	Passenger Name Record — a booking record held by the airline/GDS
OBT	Online Booking Tool
GDS	Global Distribution System (e.g. Amadeus, Sabre)
NDC	New Distribution Capability — airline direct-content standard
SSO	Single Sign-On
ERP	Enterprise Resource Planning system (finance/payroll)
Virtual card	A single-use or scoped card number issued per transaction
In-policy / out-of-policy	Whether a booking option complies with configured travel policy rules
Tenant	A client company (or business unit) using the platform, with its own employees, policies, and data, isolated from every other tenant
MoMo	Mobile Money — a telecom-operator-based digital wallet payment method (e.g.

Term	Meaning
	MTN MoMo, Airtel Money), the dominant everyday payment method in Uganda and much of East Africa
MSISDN	Mobile Station International Subscriber Directory Number — the phone number identifying a mobile money account

1.5 References

- Product Concept: Travel By MoMo — Internal Travel & Expense Platform (internal doc, this workstream)
- Business Case & Briefing (Multi-Tenant revision) — leadership-facing document covering rationale and rollout
- Booking API provider documentation (provider TBD — see Section 4.3)
- MTN MoMo / Airtel Money developer documentation (Collections and Disbursements products)
- Company travel & expense policy (existing, to be digitized as configurable rules)

2. Overall Description

2.1 Product Perspective

Travel By MoMo consists of an employee-facing application and a set of core backend services (policy engine, approval workflow, payment), shared across tenants but with data and configuration fully isolated per tenant. These services integrate with external systems: a booking-content API, a mobile money aggregator/network for the primary payment rail, a card-issuing provider for the settlement bridge, each tenant's own SSO/HR directory, and each tenant's finance/ERP system for expense sync.

2.2 Product Functions (Summary)

1. Traveler profile management
2. Trip search (flight, hotel; rail/car in later phase)
3. Booking and itinerary management
4. Policy evaluation at point of booking
5. Approval routing for out-of-policy bookings
6. Payment via Mobile Money (primary) and scoped virtual cards (settlement bridge)
7. Expense report generation and submission
8. Spend and compliance reporting
9. Notifications (booking, approval, itinerary changes)
10. Tenant administration and onboarding

2.3 User Classes and Characteristics

User class	Technical proficiency	Frequency of use
Employee/Traveler	Low–medium; expects consumer-grade booking UX and a MoMo-based reimbursement flow they already know	Occasional (per trip)
Approver/Manager	Low; needs fast decisions, not a full travel UI	Occasional, time-sensitive
Travel Admin (per tenant)	Medium–high; configures policy rules for their tenant	Regular

User class	Technical proficiency	Frequency of use
Finance (per tenant)	Medium; consumes reports, reconciles MoMo and card transactions	Regular
HR/Security (duty of care)	Low; needs traveler location in emergencies only	Rare, urgent when it happens
Tenant Admin	Medium; configures tenant-level settings (policy, approvers, branding)	Occasional, mostly at onboarding

2.4 Operating Environment

- Web application, accessible via per-tenant SSO, on desktop and mobile browsers
- Backend hosted on the company's existing cloud provider (to be confirmed based on stack)
- Must operate as a shared multi-tenant service, not a separate deployment per client company

2.5 Design and Implementation Constraints

- The system shall be designed as multi-tenant from Phase 1 onward; tenant isolation must be part of the initial data model and architecture, not retrofitted after a single-tenant build
- Must integrate with each tenant's own SSO/identity provider — no shared or independent user/password system across tenants
- Must support Mobile Money (MTN MoMo, Airtel Money) as the default, primary payment method for reimbursements and advances, given its dominant adoption among employees in the primary operating market
- Must not store raw payment card numbers (delegate to card-issuing provider) to minimize PCI scope; card rails are used only as a settlement bridge, not the primary payment method
- Must account for mobile money regulatory limits (per-transaction and daily caps set by Bank of Uganda and network operators) when sizing payment flows
- Must comply with data residency/privacy rules applicable to passport and travel-pattern data, which may vary by tenant
- Booking content is sourced via a single third-party provider initially (see Section 4.3); architecture should not hard-code assumptions that block switching providers later

2.6 Assumptions and Dependencies

- A booking-content API provider will be selected and contracted before Phase 1 engineering begins
- Each tenant's manager/approver hierarchy is available from that tenant's own HR system and is reasonably accurate
- A mobile money aggregator (e.g. Flutterwave, Pesapal) or direct MTN MoMo / Airtel Money integration can support both Collections (funding) and Disbursements (reimbursements/advances) at the volumes expected
- Mobile network/API uptime for MoMo collections and disbursements is assumed sufficient for near-real-time reimbursement and advance payments; retry/queueing logic will be needed to handle transient outages (see Section 5.3)
- A card-issuing provider relationship can support per-tenant sub-accounts or equivalent separation for the virtual-card settlement bridge (Phase 2)
- Each tenant's expense/ERP system can accept synced expense data via API or file import
- Initial tenant onboarding volume is low (single-digit tenants) and can involve some manual setup rather than fully self-serve signup

2.7 Multi-Tenancy Model

Travel By MoMo serves multiple client companies (tenants) from one shared platform rather than a separate deployment per company. Each tenant has its own employees, policy rules, approver hierarchy, and travel/expense data, fully isolated from every other tenant. Isolation approach is a foundational decision to confirm before Phase 1 design:

- Shared database, tenant_id column on every table, enforced row-level access control — lowest cost, suits many small-to-mid tenants
- Schema-per-tenant or database-per-tenant — stronger isolation, higher operational overhead, often expected by larger enterprise clients with strict compliance requirements
- A hybrid — shared by default, dedicated database for tenants who require it — is common in practice and keeps costs down without blocking larger deals

Multi-tenancy is treated as a Phase 1 requirement rather than a later addition: the core data model carries a tenant identifier from the first implementation, even while the pilot involves only one tenant. Retrofitting isolation after the data model and code assume a single tenant is substantially more expensive and riskier than building it in from the start.

3. Functional Requirements

Each requirement uses a “shall” statement and is tagged with priority (H = High, M = Medium, L = Low) and target phase, consistent with the phased rollout in the product concept (Phase 1: booking + policy visibility; Phase 2: approvals + payment; Phase 3: expense; Phase 4: reporting).

3.1 Traveler Profile Management

ID	Requirement	Priority	Phase
TP-1	The system shall authenticate users via their tenant's SSO provider.	H	1
TP-2	The system shall store traveler identity data required for booking: full legal name, date of birth, passport/ID number and expiry.	H	1
TP-3	The system shall allow travelers to store airline/hotel loyalty program numbers.	M	1
TP-4	The system shall allow travelers to set seat and meal preferences where supported by the booking provider.	L	2
TP-5	The system shall associate each traveler with a tenant, department, cost center, and default approver, sourced from that tenant's HR system.	H	1
TP-6	The system shall restrict access to traveler PII to the traveler themselves, their approver, and authorized Travel Admin/HR roles within the same tenant.	H	1

3.2 Trip Search

ID	Requirement	Priority	Phase
SR-1	The system shall allow a traveler to search for flights by origin, destination, dates, and passenger count.	H	1
SR-2	The system shall allow a traveler to search for hotels by location, check-in/check-out dates, and guest count.	H	1
SR-3	The system shall display, for each search result, whether the option is in-policy or out-of-policy for the traveler's tenant before selection.	H	1
SR-4	The system shall support rail and ground transportation search in a later phase, contingent on provider support.	L	4
SR-5	Search results shall be returned from the booking-content provider within	M	1

ID	Requirement	Priority	Phase
	5 seconds under normal load (see Section 5.1).		

3.3 Booking

ID	Requirement	Priority	Phase
BK-1	The system shall allow a traveler to select a search result and initiate a booking.	H	1
BK-2	The system shall allow a traveler to combine multiple segments (e.g. flight + hotel) into a single trip itinerary.	M	1
BK-3	The system shall capture free-text special requests/comments as part of the booking.	L	1
BK-4	The system shall issue a confirmed itinerary (or booking reference) upon successful booking with the provider.	H	1
BK-5	The system shall notify the traveler of booking confirmation via email and in-app notification.	H	1
BK-6	The system shall support cancellation/amendment of a booking, subject to the booking provider's fare rules.	M	2
BK-7	The system shall log a complete audit trail (who booked what, when, at what price, and for which tenant) for every booking.	H	1

3.4 Policy Engine

ID	Requirement	Priority	Phase
PE-1	The system shall allow a tenant's Travel Admin to configure that tenant's policy rules without requiring a code deployment.	H	1
PE-2	The system shall support, at minimum, rule types: maximum price, cabin class by flight duration, advance-booking window, and preferred vendor.	H	1
PE-3	The system shall support scoping policy rules by department, role, or seniority level within a tenant.	M	1
PE-4	The system shall evaluate every search result against the traveler's tenant's active policy rules and label it in-policy or out-of-policy before selection.	H	1
PE-5	The system shall present a human-readable reason when an option is out-of-policy (not just a flag).	M	1
PE-6	The system shall allow policy rules to be versioned per tenant, so past bookings remain evaluated against the rules in effect at booking time.	M	2

3.5 Approval Workflow

ID	Requirement	Priority	Phase
AP-1	The system shall automatically route a booking for approval if it is out-of-policy at the time of booking.	H	2
AP-2	The system shall not require approval for in-policy bookings.	H	2
AP-3	The system shall route the approval request to the traveler's designated approver within the same tenant (default: manager).	H	2

ID	Requirement	Priority	Phase
AP-4	The approver shall be able to view cost, policy violation reason, and stated trip purpose in a single view before deciding.	H	2
AP-5	The system shall support approve/reject/request-changes decisions, with an optional comment.	H	2
AP-6	The system shall escalate or notify a backup approver if a request is not actioned within a configurable time window (e.g. 24 hours).	M	2
AP-7	The system shall notify the traveler immediately of an approval decision.	H	2
AP-8	Ticketing/payment shall not proceed until the booking is approved, for any booking requiring approval.	H	2

3.6 Payment (Mobile Money-First)

Mobile Money is the primary, traveler-facing payment rail for reimbursements and advances. A virtual card, funded from the tenant's mobile-money-collected balance, is used only where a supplier requires international card-network settlement (typically global flight/hotel bookings through the content API) — travelers and tenants interact with MoMo directly wherever possible.

ID	Requirement	Priority	Phase
PM-1	The system shall support Mobile Money (MTN MoMo, Airtel Money) as the primary payment method for employee travel advances and expense reimbursements.	H	2
PM-2	The system shall use a mobile money Collections integration to receive funds when the tenant tops up its platform balance.	M	2
PM-3	The system shall use a mobile money Disbursements integration to pay travel advances and expense reimbursements directly to a traveler's registered mobile money account.	H	2
PM-4	The system shall verify a traveler's mobile money network and phone number (MSISDN) as part of traveler profile setup, required before any disbursement can be issued to that traveler.	H	1
PM-5	For booking suppliers that require international card-network settlement (most global airline/GDS and many international hotel chains), the system shall fund a scoped virtual card from the tenant's mobile-money-collected balance, used only as a settlement bridge rather than the primary traveler-facing payment method.	H	2
PM-6	The system shall not store raw payment card numbers within its own database, for any virtual cards issued.	H	2
PM-7	The system shall record and reconcile transactions across both mobile money and card-based rails against a single booking or expense, so Finance sees one unified transaction trail regardless of which rail was used.	M	2
PM-8	The system shall support configurable per-transaction and daily mobile money limits consistent with network/regulator limits, and shall split or queue payments that exceed those limits.	M	2
PM-9	The system shall support voiding/refunding a virtual card transaction if a booking is cancelled.	M	2
PM-10	The system shall log the mobile money network, transaction reference, and status (pending / completed / failed) for every mobile money transaction, tied to the associated tenant and record.	H	2

ID	Requirement	Priority	Phase
PM-11	The system shall keep payment records (mobile money and card) separated per tenant for billing and audit purposes.	H	2

3.7 Expense Management

ID	Requirement	Priority	Phase
EX-1	The system shall auto-generate a draft expense report from completed booking data (fare, hotel cost, dates).	H	3
EX-2	The system shall allow travelers to add incidental expenses (meals, ground transport, tips) with receipt upload.	H	3
EX-3	The system shall route expense reports through the same approval hierarchy used for bookings, unless a different expense-approval chain is configured for that tenant.	M	3
EX-4	The system shall sync approved expense reports to the tenant's finance/ERP system for reimbursement/payroll processing.	H	3
EX-5	The system shall flag expense line items that appear duplicated or inconsistent with the associated booking (e.g. hotel cost mismatch).	L	3
EX-6	Approved expense reimbursements shall default to mobile money disbursement, consistent with PM-3, unless a tenant configures an alternative payout method.	M	3

3.8 Reporting & Analytics

ID	Requirement	Priority	Phase
RP-1	The system shall provide a dashboard of travel spend by department, traveler, and vendor, scoped to a single tenant.	M	4
RP-2	The system shall report policy compliance rate (% of bookings in-policy) over a selectable time period, per tenant.	M	4
RP-3	The system shall report committed spend (booked but not yet traveled) separately from realized (traveled/expensed) spend.	M	4
RP-4	The system shall support exporting report data (CSV at minimum) for Finance use outside the tool, scoped to the exporting tenant only.	L	4
RP-5	The system shall report payment-rail split (mobile money vs. card) so Finance can track how much spend moves through each rail.	L	4

3.9 Notifications

ID	Requirement	Priority	Phase
NT-1	The system shall notify travelers of booking confirmations, approval decisions, and itinerary changes.	H	1–2
NT-2	The system shall notify approvers when a new approval request is pending.	H	2
NT-3	The system shall support push/email notification of flight delays or cancellations where the booking provider supplies this data.	L	4
NT-4	The system shall notify a traveler by SMS or in-app notification when a	M	2

ID	Requirement	Priority	Phase
	mobile money disbursement to them succeeds or fails, consistent with typical MoMo user expectations.		

3.10 Multi-Tenancy & Tenant Administration

Requirements governing tenant isolation, tenant-scoped configuration, and tenant administration, introduced as a foundational Phase 1 concern per Section 2.7.

ID	Requirement	Priority	Phase
MT-1	The system shall associate every persisted record (traveler, trip, booking, policy, approval, payment, expense) with exactly one tenant identifier.	H	1
MT-2	The system shall enforce tenant-scoped data access at the data-access layer, such that no query can return records belonging to a different tenant, regardless of application-layer logic.	H	1
MT-3	The system shall support tenant-specific policy rule sets, configurable independently per tenant.	H	1
MT-4	The system shall support tenant-specific approver hierarchies, sourced from each tenant's own HR/identity system.	H	2
MT-5	The system shall support federated SSO per tenant, routing authentication to the correct tenant's identity provider.	H	1
MT-6	The system shall provide a Tenant Admin role scoped to a single tenant, permitted to configure that tenant's policy rules, approver hierarchy, and branding, with no visibility into other tenants.	M	1
MT-7	The system shall support onboarding a new tenant (tenant record, initial admin user, default policy set) without requiring a code deployment.	M	2
MT-8	The system shall scope all reporting and analytics to a single tenant by default, with no cross-tenant aggregation exposed to tenant-level users.	H	4
MT-9	The system shall log all administrative actions with the acting tenant and user identified, for audit purposes.	M	1

4. External Interface Requirements

4.1 User Interfaces

- Responsive web application; primary interaction is search → results → booking
- Approvals accessible from a simplified mobile-friendly view (approvers should not need the full app to approve/reject)
- Policy status (in-policy/out-of-policy) must be visible before the traveler commits to a selection, not only after
- Reimbursement/advance status shown in terms travelers already recognize from personal MoMo use (e.g. pending, completed, failed with a reason)

4.2 Hardware Interfaces

Not applicable — Travel By MoMo is a standard web application with no dedicated hardware dependency.

4.3 Software Interfaces

Interface	Purpose	Notes
Booking-content API (provider TBD, e.g. Duffel / Amadeus Self-Service APIs / Spotnana)	Flight/hotel search, booking, cancellation	Selection should happen via a short bake-off against real travel patterns before Phase 1 build
Mobile Money aggregator (e.g. Flutterwave, Pesapal) or direct MTN MoMo / Airtel Money APIs	Primary payment rail: Collections (tenant funding) and Disbursements (traveler reimbursements/advances)	Preferred default; an aggregator covers both major Uganda networks through one integration rather than integrating each network separately
Card-issuing provider (e.g. Stripe Issuing, Marqeta)	Funds a scoped virtual card from the mobile-money-collected balance, used only where a supplier requires international card-network settlement	Secondary settlement bridge, not the primary traveler/company-facing payment method; Phase 2 dependency
Tenant SSO / identity providers	Authentication, tenant identification, role/department data	Each tenant federates its own IdP; the platform must support multiple concurrent identity providers, not a single fixed one
Tenant HR systems	Manager hierarchy, department/cost center data, per tenant	Source of truth for approver routing within each tenant
Tenant finance/ERP systems	Expense sync for reimbursement/payroll, per tenant	Phase 3 dependency; integration method (API vs. file) may vary by tenant

4.4 Communications Interfaces

- HTTPS/TLS for all client-server and server-to-provider communication
- Webhook or polling support required from the booking-content provider for itinerary change notifications (Phase 4)
- Webhook or polling support required from the mobile money aggregator/network to receive Collection and Disbursement status updates

5. Non-Functional Requirements

5.1 Performance

- Search results shall be returned within 5 seconds under normal load (95th percentile)
- Booking confirmation shall complete within 10 seconds of traveler submission under normal load
- A mobile money disbursement request shall be submitted to the aggregator/network within 5 seconds of approval, though final settlement time depends on the network

5.2 Security

- No raw payment card data stored in the system's own database (delegated to card-issuing provider)
- Mobile money account identifiers (MSISDN) treated as sensitive data, with access restricted equivalently to payment card data
- Traveler PII (passport data, DOB) encrypted at rest and in transit
- Role-based access control enforced for all traveler and financial data, scoped within tenant boundaries
- All administrative actions (policy changes, approval overrides, manual payment retries) logged with actor, tenant, and timestamp

5.3 Availability & Reliability

- Target uptime: 99.5% during business hours, given this is not a 24/7 mission-critical system, but travel disruptions can occur outside business hours — booking-provider outage handling should degrade gracefully, not silently fail
- Booking and payment operations shall be idempotent to avoid duplicate charges/bookings/disbursements on retry
- Mobile money collection and disbursement calls shall retry with backoff and queue rather than fail silently if the aggregator/network is temporarily unavailable

5.4 Scalability

System shall support the current and reasonably projected number of tenants and employees per tenant, and the associated mobile money transaction volume, without redesign (specific targets to be confirmed with stakeholders).

5.5 Maintainability

- Policy rules configurable by non-engineers (Travel Admin) without deployment
- Booking-content provider integration isolated behind an internal adapter layer to allow provider replacement without rewriting core services
- Payment rail integrations (mobile money aggregator, card issuer) isolated behind a payment adapter layer so either can be swapped without rewriting booking or expense logic

5.6 Compliance & Legal

- PCI DSS scope minimized by delegating card handling to the card-issuing provider; PCI DSS applies specifically to the card-rail settlement bridge, not to mobile money transactions
- Mobile money transactions are subject to Bank of Uganda's National Payment Systems Act and related e-money/agent regulations, and to the relevant regulator in any other market a tenant operates in
- Data privacy handling for passport/ID data consistent with applicable regulations (e.g. GDPR if EU employees/travel or EU tenants are in scope) — data residency requirements to be confirmed with Legal, and may vary by tenant
- Audit trail retained for all bookings, approvals, and payment transactions for a period consistent with each tenant's financial record-keeping policy

5.7 Usability & Accessibility

- Core booking flow should be completable without training or a user manual
- Reimbursement/advance flow should feel familiar to anyone who already uses MoMo personally — no new payment concepts to learn
- Web interface shall meet the company's existing internal accessibility standard (e.g. WCAG 2.1 AA if that is the baseline)

5.8 Multi-Tenancy & Isolation

- Tenant data isolation shall be enforced at the data-access layer, not solely through application logic
- Automated tests shall verify tenant isolation by attempting cross-tenant access and confirming it is denied, run as part of continuous integration
- The chosen tenant isolation approach (Section 2.7) shall be documented and reviewed by Engineering leadership and Legal before Phase 1 sign-off
- A security incident involving cross-tenant data exposure shall be treated as a severity-1 incident regardless of the number of records affected

6. Data Requirements

6.1 Core Entities

tenants (tenant identifier, name, status, isolation tier), travelers (including mobile money network + MSISDN), trips, bookings (with segments: flight/hotel/car), policies (versioned rule sets, tenant-scoped), approvals, payments (mobile money and card, both rails represented in one model), expenses. Every entity below the tenants table itself carries a tenant identifier.

6.2 Data Retention & Privacy

- Traveler PII retained only as long as required for active employment and legal/financial record-keeping requirements
- Passport/ID data access logged and restricted to booking and compliance purposes only
- Mobile money account identifiers (MSISDN) retained only as long as needed to issue reimbursements/advances, and access-logged the same way as passport data
- Tenant data shall not be commingled in exports, backups, or analytics outputs accessible to tenant-level users

7. Appendix: Requirement Priority Summary

Priority	Meaning
H (High)	Required for the phase to deliver its core value; blocks phase sign-off
M (Medium)	Materially improves the tool; can slip to the next phase under schedule pressure
L (Low)	Nice-to-have; explicitly deferrable without undermining the phase's goal

This is a draft for internal scoping. Sections 4.3 (provider selection), 2.7/5.8 (tenant isolation approach), 3.6/4.3 (mobile money aggregator vs. direct network integration), and 5.6 (compliance specifics) should be finalized with the relevant stakeholders (Engineering leadership, Legal, Finance) before Phase 1 design begins.